

Yuri Shiroiva Berbereia

AI Engineer / Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist with 3+ years of experience building generative AI systems and data pipelines in production. At IoTa, I architected a multi-agent LLM system that cut reverse-engineering time for CAN bus/ISOBUS telemetry in half, shipping 20+ decoded signals to production, and an LLM diagnostics feature that reduced analysis time from 2 weeks to 5 days. I work with Python, SQL, GCP/Vertex AI, Cassandra, PostgreSQL, and PyTorch. Technologist Degree in Applied Artificial Intelligence from PUCPR, with honorable mention.

TECHNICAL SKILLS

Generative AI & LLMs: production LLM applications, multi-agent systems, Claude on Vertex AI, MCP (Model Context Protocol), prompt engineering, structured output with JSON Schema, RAG, LangChain, Hugging Face Transformers

Machine Learning: PyTorch, scikit-learn, gradient boosting, CNNs, semantic segmentation (U-Net, U-Net++, DeepLabV3+), ensembles, multi-task learning, cross-validation, bootstrap confidence intervals

Data: Python, SQL, pandas, NumPy, ETL pipelines, data quality, Apache Cassandra, PostgreSQL, bi-temporal data modeling, time series, geospatial data (GeoJSON, Sentinel-2)

Cloud & Tools: GCP (Vertex AI, IAM, VPC), Docker, Git, REST APIs, Streamlit

Domain: agricultural machinery telemetry, CAN bus / ISOBUS (J1939, ISO 11783, ISOXML), precision agriculture

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer — IoTa (IoT platform for AgTech) 2024 – Present

- Cut reverse-engineering time for undocumented CAN messages in half, from 1 month to 2 weeks per machine model, by architecting a multi-agent LLM system that orchestrates parallel Python kernels through MCP servers, with ground-truth validation and hold-out testing; 20+ decoded signals are now in production.
- Reduced agricultural machinery fault diagnosis time from 2 weeks to 5 days by designing end to end an LLM feature (Claude on Vertex AI) that summarizes Cassandra telemetry and delivers schema-validated structured diagnostics to the data science team.
- Expanded platform support to 10+ equipment brands and models by implementing a plugin registry for J1939/ISOBUS decoding that isolates each brand from the pipeline core.
- Unlocked geospatial data for 200+ fields locked in a vendor's proprietary format by integrating that format into the data pipeline, with no dependency on the partner's API.
- Replaced manual assembly of one ISOXML file per operation per machine with a single template generated automatically by a Python pipeline, along with GeoJSON coverage maps built from raw CAN bus telemetry.
- Ensured full traceability of every field operation's configuration by designing an append-only bi-temporal schema in Cassandra, allowing the state of any operation to be reconstructed at any point in its history.
- Fixed applied-product totals reported to customers by diagnosing two production defects in the ingestion service: a unit conversion error in flow delay compensation and an aggregation scope bug.

Software Development Intern — Optcore 04/2023 – 05/2024

- Evolved the Streamlit and Appsmith dashboards a client company's leadership used to track results across its portfolio of about 30 customers.
- Built Python data processing for shopping mall data (cleaning, standardization, and deduplication across different sources) for a retail analytics project at prototype stage.

PROJECT

Field Boundary Segmentation with Deep Learning (Undergraduate Thesis) — PUCPR 2026

- Outperformed the semantic baseline on the FTW-Brazil benchmark in Panoptic Quality (0.51 vs. 0.46) for field segmentation in Sentinel-2 imagery, using a weighted ensemble of 6 networks (U-Net, U-Net++, and DeepLabV3+ with ResNet-34/50 encoders) and TTA, reaching a mean Dice of 0.77 (median 0.87).
- Deployed the solution at IoTa as a university extension project, where 5 professionals from engineering, data science, and DevOps rated it 4.6/5 on average for usefulness and accuracy.
- Reached a mean precision of 0.79 by calibrating the decision threshold and removing spurious components in post-processing, and validated results with 95% confidence intervals and effect size (Cohen's d).

EDUCATION

- **Technologist Degree in Applied Artificial Intelligence** — PUCPR | 2024 – 2026. GPA 9.27/10. Honorable mention for overall academic performance.
- **Electrical Engineering (Robotics and Industrial Automation)** — UFJF | 2021 – 2023. Not completed; transferred to PUCPR in 2024.

CERTIFICATIONS & LANGUAGES

Certifications: IBM Generative AI Engineering (2026) · IBM Generative AI Engineering with LLMs (2026) · Supervised Machine Learning, DeepLearning.AI (2025) · IBM Data Science Professional (2024) · Google Data Analytics (2023)

Languages: Portuguese (native) · English (proficient technical reading, basic conversation) · Spanish (basic)